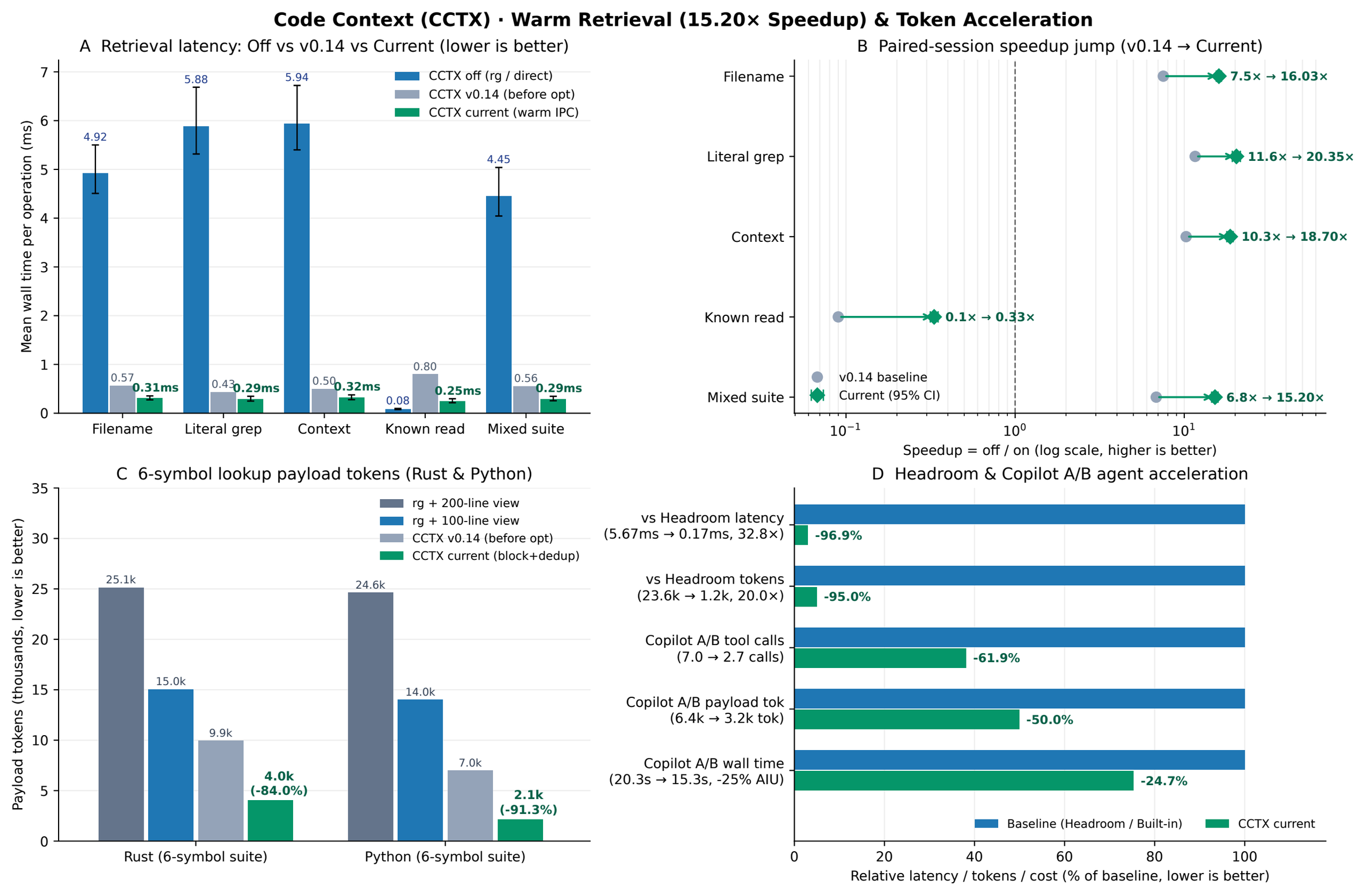
Code Context (CCTX) · Warm Retrieval (15.20× Speedup) & Token Acceleration
A Retrieval latency: Off vs v0.14 vs Current (lower is better)
Mean wall time per operation (ms)
CCTX off (rg / direct)
CCTX v0.14 (before opt)
CCTX current (warm IPC)
4.92
0.57
0.31ms
5.88
0.43
0.29ms
5.94
0.50
0.32ms
0.08
0.80
0.25ms
4.45
0.56
0.29ms
Filename
Literal grep
Context
Known read
Mixed suite
B Paired-session speedup jump (v0.14 → Current)
Filename
Literal grep
Context
Known read
Mixed suite
7.5× → 16.03×
11.6× → 20.35×
10.3× → 18.70×
0.1× → 0.33×
6.8× → 15.20×
v0.14 baseline
Current (95% CI)
Speedup = off / on (log scale, higher is better)
C 6-symbol lookup payload tokens (Rust & Python)
Payload tokens (thousands, lower is better)
rg + 200-line view
rg + 100-line view
CCTX v0.14 (before opt)
CCTX current (block+dedup)
25.1k
15.0k
9.9k
4.0k
(-84.0%)
24.6k
14.0k
7.0k
2.1k
(-91.3%)
Rust (6-symbol suite)
Python (6-symbol suite)
D Headroom & Copilot A/B agent acceleration
vs Headroom latency
(5.67ms → 0.17ms, 32.8×)
-96.9%
vs Headroom tokens
(23.6k → 1.2k, 20.0×)
-95.0%
Copilot A/B tool calls
(7.0 → 2.7 calls)
-61.9%
Copilot A/B payload tok
(6.4k → 3.2k tok)
-50.0%
Copilot A/B wall time
(20.3s → 15.3s, -25% AIU)
-24.7%
Baseline (Headroom / Built-in)
CCTX current
Relative latency / tokens / cost (% of baseline, lower is better)